PAYTM BUILD FOR INDIA AI HACKATHON · CHENNAI · TRACK 3

Merchant Support Resolver

An autonomous AI teammate that diagnoses and fixes merchant issues, so merchants stop waiting on tickets.

Team Fixr · Autonomous AI Teammates

PROBLEM STATEMENT

Merchants lose money and time waiting on support

Settlement not received

Cash flow stalls while the merchant chases a UTR and a status nobody can see.

QR or Soundbox down

Every hour offline is lost sales and customers who pay in cash or walk away.

Chargeback or dispute

Unclear next steps and deadlines, so evidence is submitted late or not at all.

Each ticket today: explain, wait for a human to look up the data, then wait again for the fix.

Scale: Paytm’s merchant network spans 1.51 crore storefronts (Paytm, May 2026).

PROPOSED SOLUTION

An AI teammate that resolves the ticket end to end

Reads and remembers

Reads the message in Hindi, Tamil or English, pulls the merchant's data, and keeps the context of earlier turns.

Decides with guardrails

One policy gate in code approves every read and every action, with limits on retries and tickets.

Takes action

Re-triggers a settlement, raises a bank ticket, restarts a device, submits dispute proof, or hands the case to a human.

Replies in the merchant’s language: Hindi, Tamil and English. Voice is the next step.

PROPOSED SOLUTION · HOW IT WORKS

One ticket, four steps, no handoffs

1

Understand

Reads the message and pulls merchant data, with earlier turns.

2

Diagnose

Finds the root cause using tools and rules.

3

Act

Runs the fix inside policy limits, or escalates with a case file.

4

Report

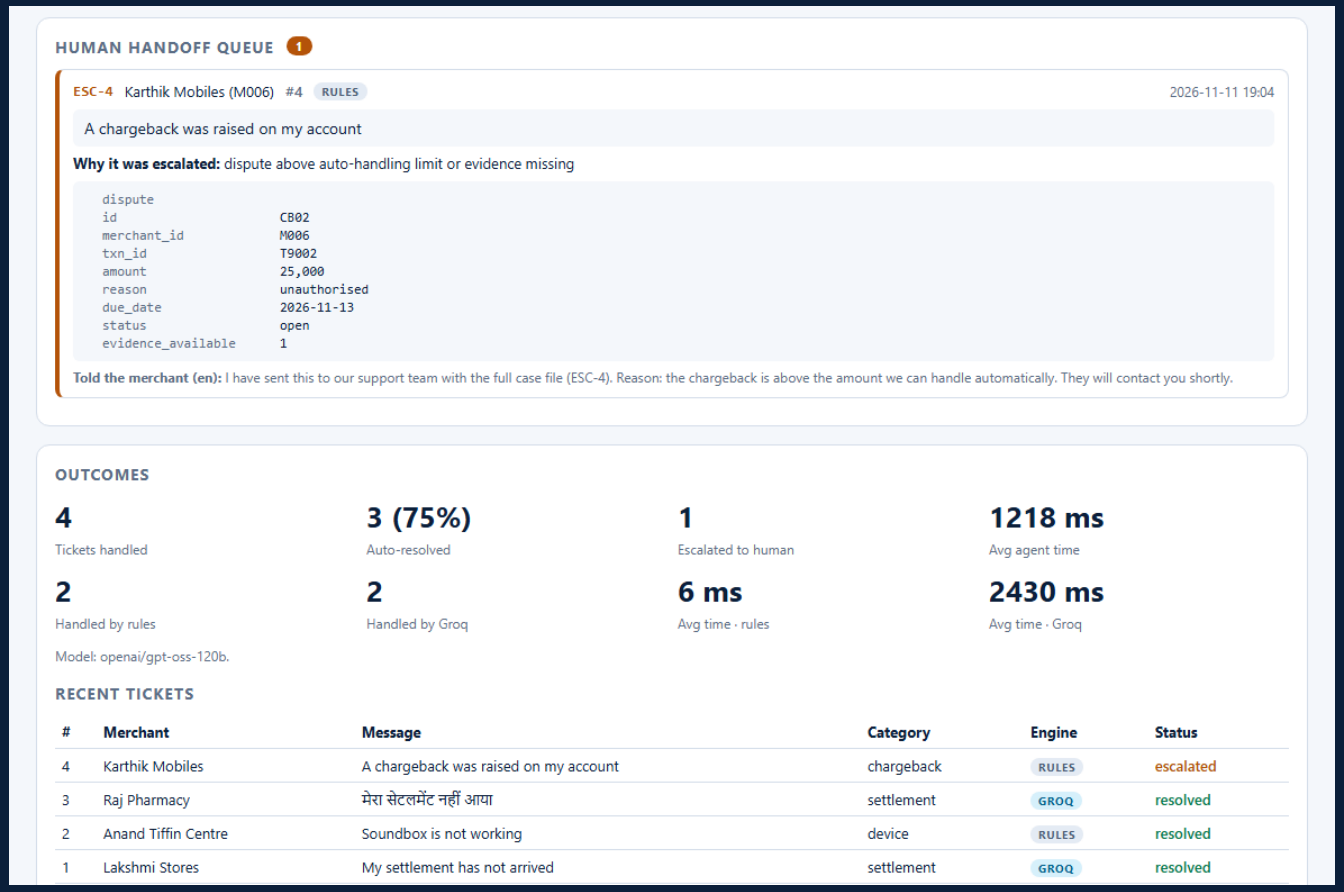
Tells the merchant what was done, in their own language.

Example

“Mera settlement nahi aaya” → settlement found, pending at the bank → bank ticket raised → merchant told in Hindi: expected credit tomorrow, 11 Nov.

LIVE DEMO

The whole loop, visible in one screen



HUMAN HANDOFF QUEUE 1

ESC-4 Karthik Mobiles (M006) #4 RULES 2026-11-11 19:04

A chargeback was raised on my account

Why it was escalated: dispute above auto-handling limit or evidence missing

```
dispute
id                  CB02
merchant_id         M006
txn_id              T9002
amount              25,000
reason              unauthorised
due_date            2026-11-13
status              open
evidence_available  1
```

Told the merchant (en): I have sent this to our support team with the full case file (ESC-4). Reason: the chargeback is above the amount we can handle automatically. They will contact you shortly.

OUTCOMES

4 Tickets handled	**3 (75%)** Auto-resolved	**1** Escalated to human	**1218 ms** Avg agent time
2 Handled by rules	**2** Handled by Groq	**6 ms** Avg time · rules	**2430 ms** Avg time · Groq

Model: openai/gpt-oss-120b.

RECENT TICKETS

#	Merchant	Message	Category	Engine	Status
4	Karthik Mobiles	A chargeback was raised on my account	chargeback	RULES	escalated
3	Raj Pharmacy	मेरा सेटलमेंट नहीं आया	settlement	GROQ	resolved
2	Anand Tiffin Centre	Soundbox is not working	device	RULES	resolved
1	Lakshmi Stores	My settlement has not arrived	settlement	GROQ	resolved

Human handoff queue

Every case the agent could not fix inside policy, with the case file and what the merchant was told.

Outcomes, measured

Tickets handled, auto-resolved, escalated, average time — and the rules-vs-Groq split on the same run.

Engine on every ticket

Which engine answered, which model, and whether it fell back. Two engines, one audit log.

TECHNOLOGY STACK

Groq for reasoning, plain Python for safety

Layer	Choice
Agent brain	Groq (groq.com) tool calling on openai/gpt-oss-120b, with a free rule-based fallback and automatic retry when the API rate-limits
Orchestration	Plain Python loop; every read and every action passes one policy gate in code
Data and tools	SQLite with synthetic data; 9 tools to investigate, act and escalate
Channels	Web chat in Hindi, Tamil and English; voice is a next step
Guardrails	One gate for reads and actions, action caps, human handoff, full audit log
Quality	115 automated tests, CI on Python 3.9–3.13, and a 10-finding audit of the project closed to zero open

Built to plug into Paytm’s real settlement, device and ticketing APIs; the demo uses synthetic data.

SAFETY BY DESIGN

The model proposes. The code disposes.

One checkpoint

policy_gate() in code is the only thing that can approve a read or an action.

Bounded by limits

At most 2 retriggers, 2 bank tickets and 2 restarts per record. A refused action escalates.

No case left unhandled

If the model only describes a case, the rule engine decides and a human still gets it.

Scoped to the merchant

merchant_id is injected in code, a cross-merchant read is refused, and every call is written to the audit log the support team reads.

UNIQUE SELLING PROPOSITION

Resolves, not just replies

Three differences a merchant notices in the first minute: the case closes itself, the reply arrives in their own language, and nothing happens outside policy.

Closes the loop

Diagnoses the cause and executes the fix through tools, instead of pasting an FAQ link.

Speaks their language

Replies in Hindi, Tamil and English, and remembers the conversation, so a follow-up does not start over.

Safe by design

Every read and every action passes one gate in code, with limits and a human handoff. Nothing outside policy happens.

IMPACT AND BENEFITS

Faster fixes for merchants, fewer tickets for humans

Merchants

Fixes in minutes, any hour, in their own language, with a clear next step.

Paytm

Lower cost per ticket, better merchant retention and trust as the network grows.

Support team

Agents spend time on hard cases, with the diagnosis already done.

What we will measure

First-contact resolution · time to resolution · escalation rate · merchant satisfaction. Demo run: 4 tickets handled, 3 auto-resolved (75%), 1 escalated with a case file, 2 by Groq and 2 by rules, 0 fallbacks.

BUSINESS MODEL

Start with cost savings, grow into a merchant product

Phase 1

Pilot

Resolve the top three ticket types inside Paytm's support desk. Value: lower cost per ticket.

Phase 2

Merchant-facing

Offer the teammate inside the Paytm for Business app, with voice in regional languages.

Phase 3

Expand

Add proactive nudges and sales or lending follow-ups on the same agent platform.

> **Revenue levers: support cost saved, a premium priority-resolution tier for merchants, and higher retention.**